

AAE 451B – A6: Propulsion System

Team 01

9/20/26



School of Aeronautics
and Astronautics

Propulsion Requirements

- Electrical power: 406 W per power plant, 812 W total
 - From selected power loading and vehicle weight estimate
 - Assumptions:
 - Cruise throttle = 0.75
 - $\eta_{\text{prop,cruise}} = 0.75$
 - $\eta_{\text{prop,TO}} = 0.45$
 - $\eta_{\text{motor}} = 0.80$
 - $\eta_{\text{ESC}} = 0.95$
- Static thrust: 15.24 N
- Cruise thrust: 5.59 N

Component Selection

Configuration	Motor	Propeller	ESC	Battery	Pack Voltage	Motor Spec
1	C-2820/10	APC 8x4-E	RC Electric Parts Brushless ESC Classic	Turnigy 4S 4000 mAh 30C	14.8 V	Link
2	C-2814/12	APC 7x4-E	RC Electric Parts Brushless ESC Classic	Turnigy 4S 4000 mAh 30C	14.8 V	Link
3	C-2221/10	APC 7x6-E	RC Electric Parts Brushless ESC Classic	Common Sense RC 5200 mAh 50C	11.1 V	Link
4	C-2814/10	APC 7x6-E	RC Electric Parts Brushless ESC Classic	Common Sense RC 5200 mAh 50C	11.1 V	Link

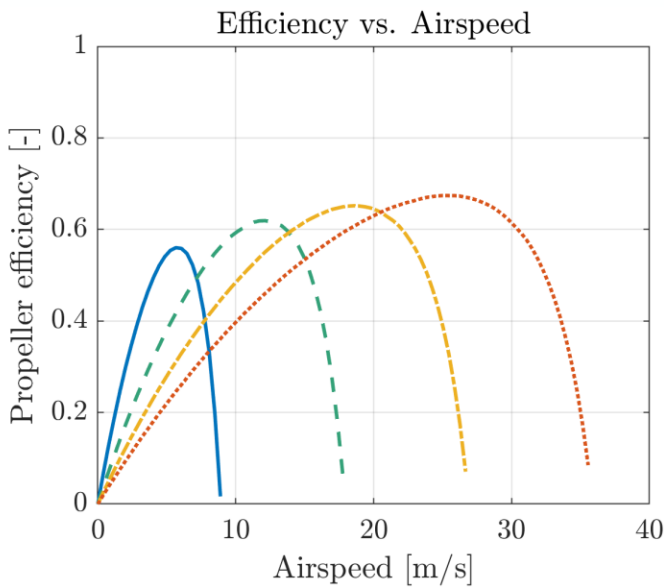
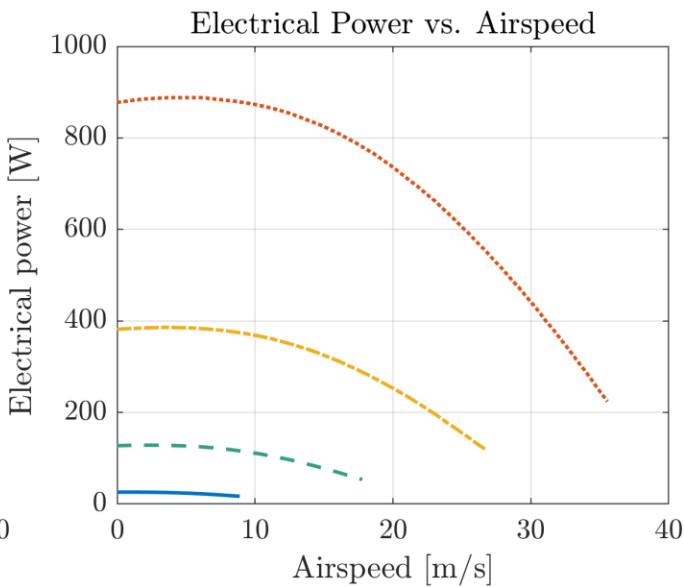
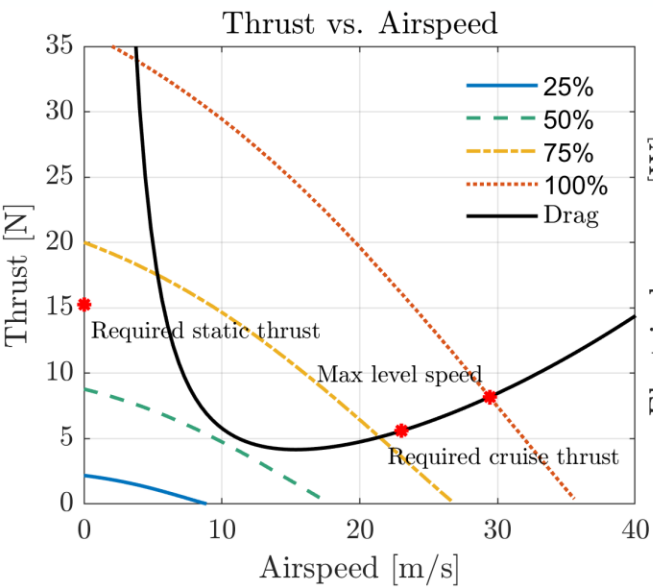
Propulsion Performance Analysis

- Propeller data from APC
- Data values linearly interpolated from RPM and airspeed/advance ratio inputs
- Full throttle RPM determined by checking torque balance and propeller structural limits
 - Minimum of the two values sets maximum RPM
- Assumed RPM linear with throttle

Config 1

- Meets all criteria except weight budget
- Peak efficiency of about 67%

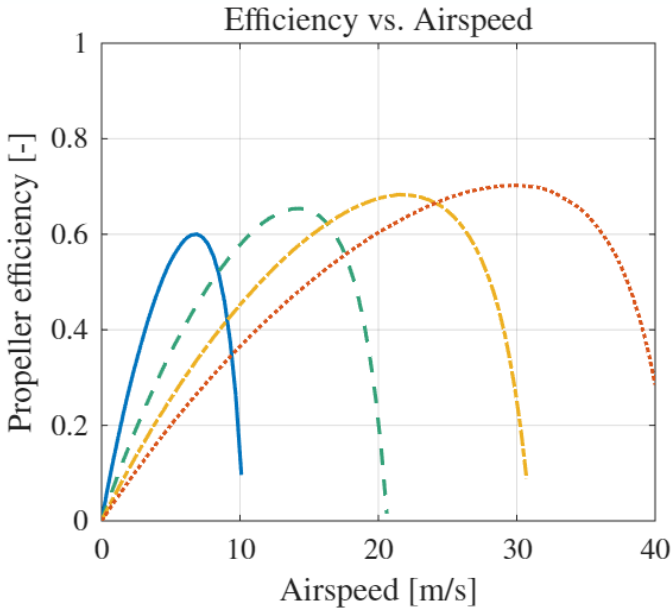
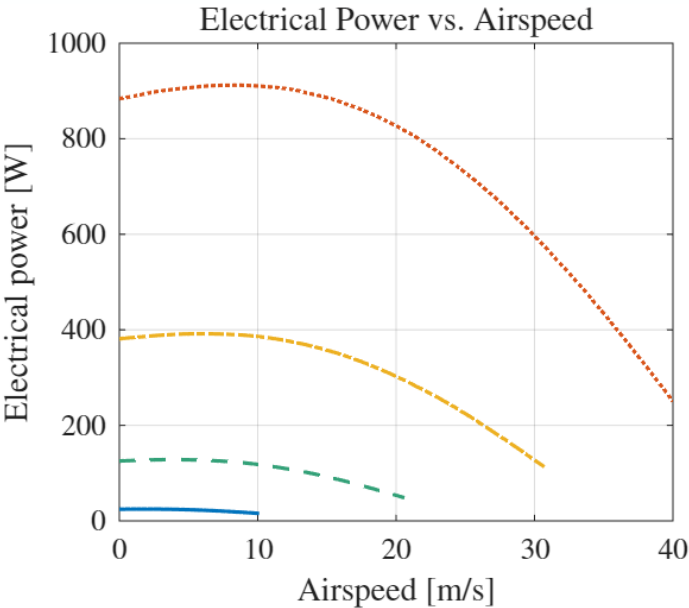
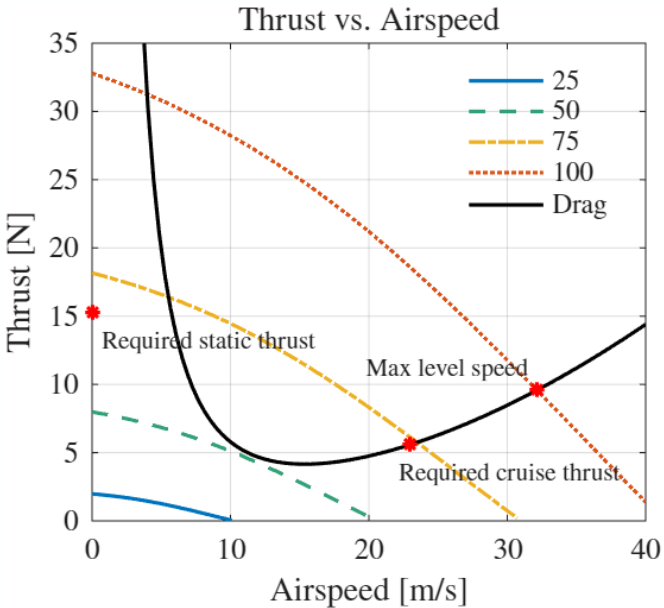
Factor	Value	Meets Needs?
Motor Input Voltage Range	11.1-18.5 V	Y
Motor Max Cont. Current	48 A	Y
Motor Kv	1170	Y
Motor Mass	142 g	Y
Battery Mass	452 g	N
Prop D, Ground Clearance	8 in D, 2.5 in Clearance	Y
ESC Rating – Max Motor Current	12 A	Y
Battery Cells, Discharge Rate	4 Cells, 30 C Discharge	Y
Connectors	XT60	Y



Config 2

- Meets all criteria except weight budget
- Peak efficiency of about 70%

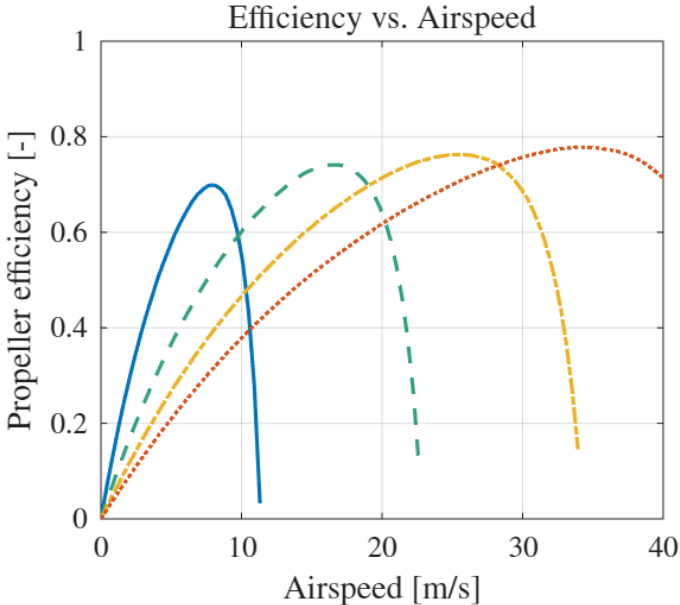
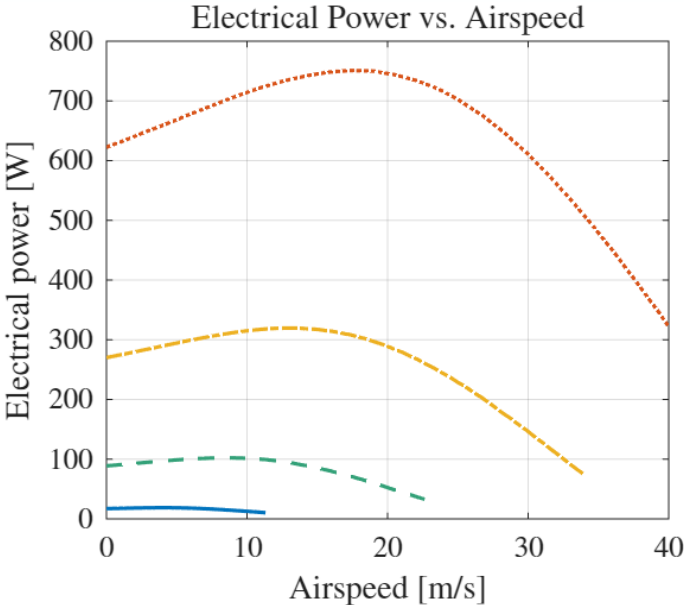
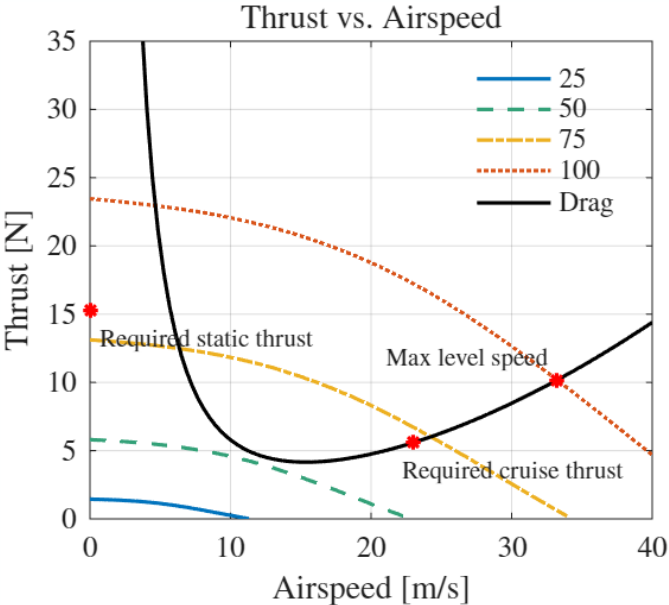
Factor	Value	Meets Needs?
Motor Input Voltage Range	11.1-18.5 V	Y
Motor Max Cont. Current	40 A	Y
Motor Kv	1390	Y
Motor Mass	109 g	Y
Battery Mass	452 g	N
Prop D, Ground Clearance	7 in D, 3.5 in Clearance	Y
ESC Rating – Max Motor Current	20 A	Y
Battery Cells, Discharge Rate	4 Cells, 30 C Discharge	Y
Connectors	XT60	Y



Config 3

- Meets all criteria
- Peak efficiency of about 78%

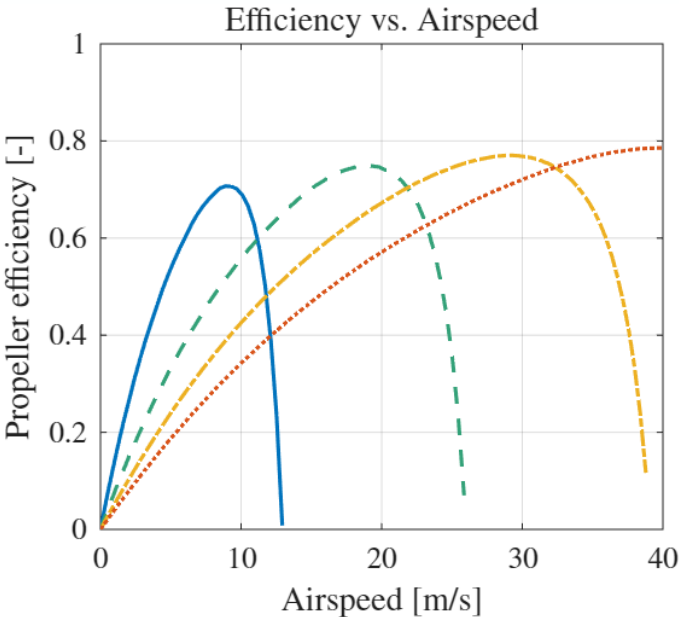
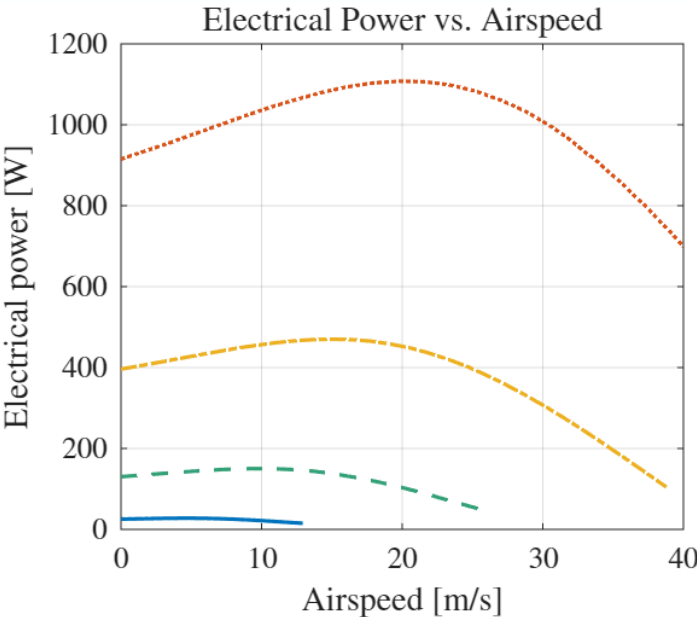
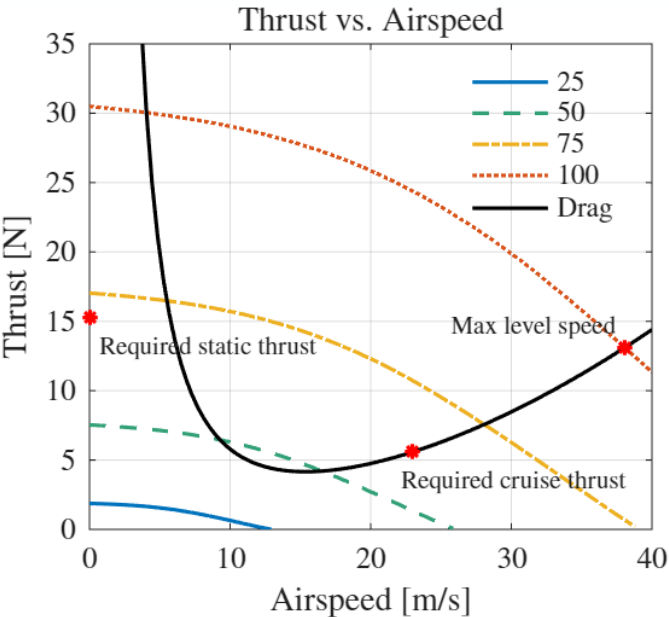
Factor	Value	Meets Needs?
Motor Input Voltage Range	11.1-14.8 V	Y
Motor Max Cont. Current	38 A	Y
Motor Kv	1500	Y
Motor Mass	88 g	Y
Battery Mass	350 g	Y
Prop D, Ground Clearance	7 in D, 3.5 in Clearance	Y
ESC Rating – Max Motor Current	22 A	Y
Battery Cells, Discharge Rate	3 Cells, 50 C Discharge	Y
Connectors	XT60	Y



Config 4

- Meets all criteria except RFP power limit
- Peak efficiency of about 79%

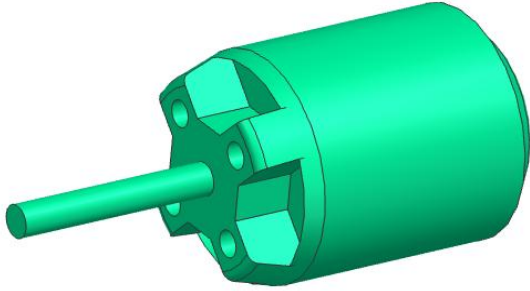
Factor	Value	Meets Needs?
Motor Input Voltage Range	11.1-18.5 V	Y
Motor Max Cont. Current	48 A	Y
Motor Kv	1700	Y
Motor Mass (each)	109 g	Y
Battery Mass	350 g	Y
Prop D, Ground Clearance	7 in D, 3.5 in Clearance	Y
ESC Rating – Max Motor Current	12 A	Y
Battery Cells, Discharge Rate	3 Cells, 50 C Discharge	Y
Connectors	XT60	Y



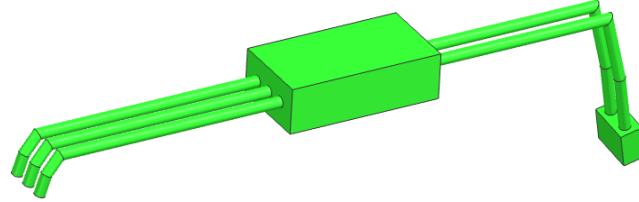
System Summary

Component	Selected Part	Cost	Mass
Motor	C-2221/10 (x2)	\$65.98	176 g
Propeller	APC 7x6-E (x2)	\$5.42	9 g
ESC	RC Electric Parts Brushless ESC Classic (x2)	\$56.98	136 g
Battery	Common Sense RC 5200 mAh 50C	\$72.09	350g
Totals	Config 3 Selected	\$200.47 (\$215.20 after tax)	671 g

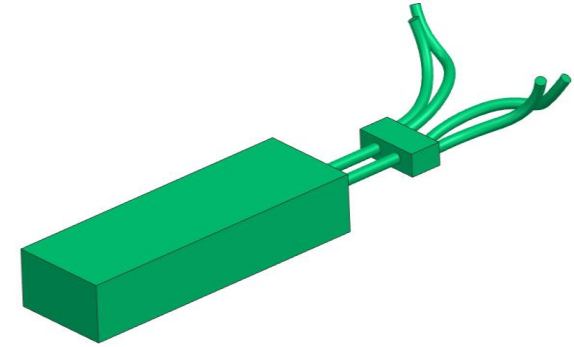
CAD Model - Components



Motor
(C-2221/10)



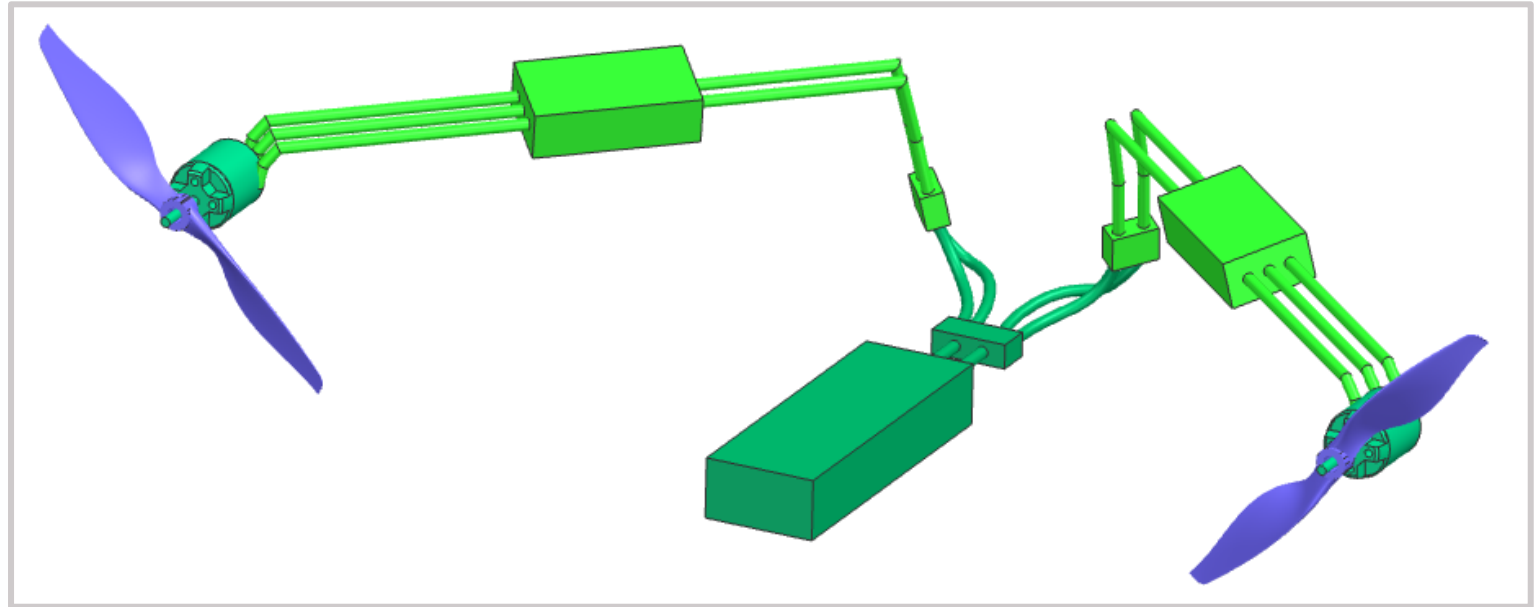
ESC
(RC Electric Parts Brushless ESC Classic)



Battery
(Common Sense RC 5200 mAh 50C)



Propeller
(APC 7x6-E)

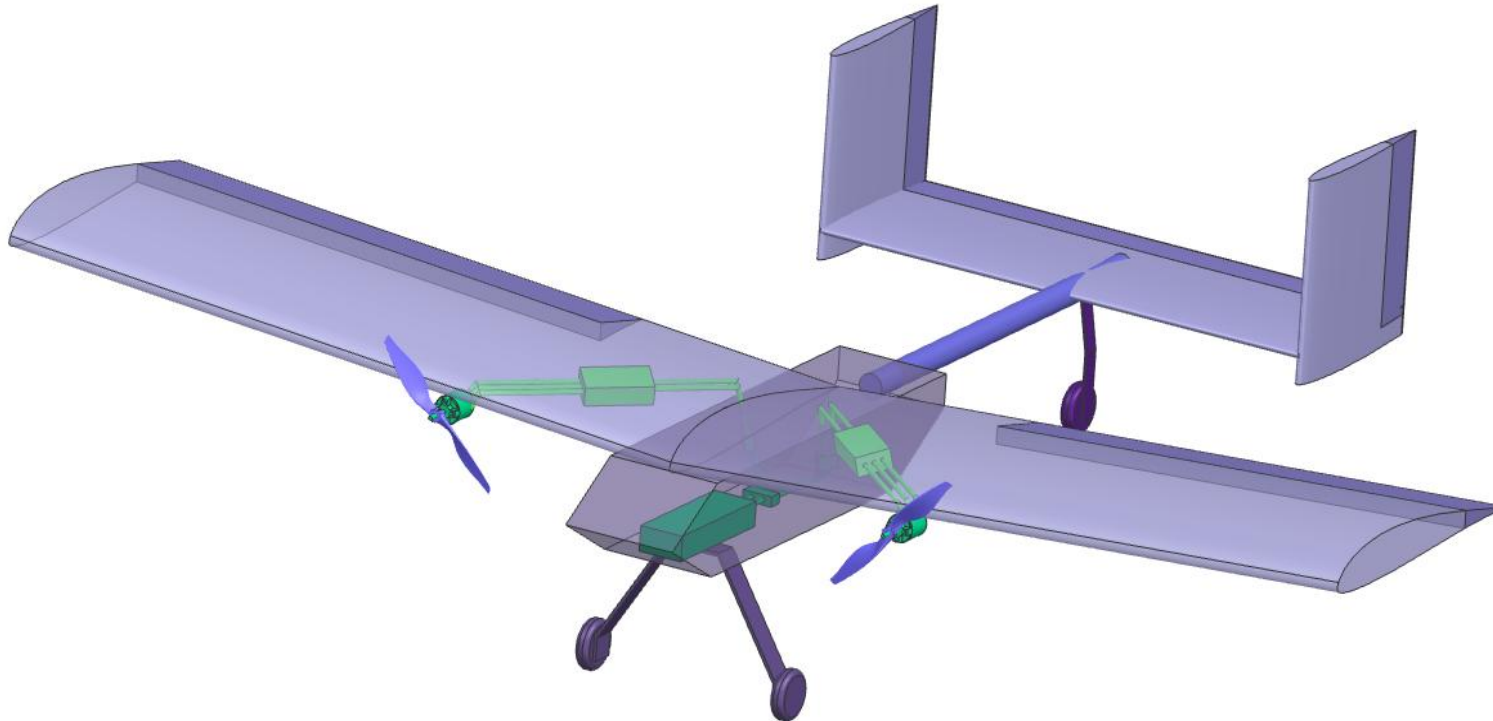


Propulsion Assembly

CAD Model – Updated Assembly

- Updated assembly model is in a breakout file from the parameterized model, and lives inside the team folder. It is titled "**AAE45100 F26 – Team 01 Configuration Assembly**" as shown below.

☐	 AAE_451_Parametric_Model_Student_Manual_DRAFT	Aug 20, 2026	Document	In Work	Christos Koumtzis	▼
☐	 AAE45100 F26 - Team 01 Configuration Assembly	Sep 20, 2026	Physical Product	In Work	Sarah Vose	▼
☐	 AAE45100 F26 - Team 01 Configuration Assembly	Sep 20, 2026	3D Shape	In Work	Sarah Vose	▼
☐	 AAE 451 Teamwork	Sep 17, 2026	Bookmark Root	Draft	Kevin Stevens	▼



Updated Full
Assembly Model